

Figure 17: Physical Validation — 3D-Print Test Protocol (Future Work)

(a) Scaled 3D-Print Model

**PHOTO
PLACEHOLDER**

Physical prototype required

1:20 scale model of optimized
SASTO-PA geometry, fabricated via
structural concrete 3D printing.

Material: Structural concrete
Print volume: ~50 x 45 x 35 cm
Layer height: 1.5 mm

Status: FUTURE WORK

(b) Load Test Setup

**PHOTO
PLACEHOLDER**

Compression test required

Universal testing machine with
distributed loading plate.
DIC (Digital Image Correlation)
for full-field strain measurement.

Protocol: ASTM C39 adapted
Loading rate: 0.25 MPa/s

Status: FUTURE WORK

(c) Failure Analysis

**PHOTO
PLACEHOLDER**

Post-failure inspection required

Compare failure mode and load
with FEA predictions:

- Failure load vs. predicted
- Crack pattern vs. stress map
- Deformation vs. displacement

Acceptance: Within 20% of FEA

Status: FUTURE WORK